Replication Guide

Bug Report Classification Tool — How to Reproduce Results

Step 1: Clone the Repository

```
git clone https://github.com/sheunsho/ISE.git
cd ISE
```

Step 2: Set Up the Environment

```
python -m venv venv
source venv/bin/activate      # macOS/Linux
pip install -r requirements.txt
```

Step 3: Verify the Datasets

Confirm the following CSV files exist in `lab1/datasets/`:

- `tensorflow.csv`
- `pytorch.csv`
- `keras.csv`
- `incubator-mxnet.csv`
- `caffe.csv`

These datasets are included in the repository. They originate from the ISE course GitHub repository (https://github.com/ideas-labo/ISE).

Step 4: Run the Experiment

```
cd lab1
python main.py
```

A single invocation of `main.py` runs two experiments back-to-back:

default — improved classifiers (SVM, Random Forest, Logistic Regression, XGBoost) run with no class imbalance handling. The Naive Bayes baseline retains its uniform class prior in both modes, since this prior is a baseline correction (not an imbalance-handling technique) and without it the baseline scores F1 near zero.

weighted — the main experiment reported in the paper. Improved classifiers use class weighting (`class_weight='balanced'` for SVM, Random Forest, Logistic Regression; `scale_pos_weight` equal to the negative/positive ratio for XGBoost).

Expected runtime: approximately 4–10 minutes total (both modes combined). The script prints progress for each project as it runs.

Step 5: Verify the Results

After the script completes, check that the following files have been created. Files from the main (weighted) experiment keep unsuffixed names so they match the figure references in the LaTeX report. Files from the default experiment are suffixed with `_default` so both sets coexist.

Weighted experiment (main results):

- `results/raw_results.csv — 750 rows of individual run metrics`
- `results/summary_results.csv — 25 rows of aggregated statistics`
- `results/wilcoxon_tests.csv — statistical test outcomes`

- `results/figures/{project}_f1_boxplot.png — one box plot per project (5 files)`

Default experiment (for comparison):

- `results/raw_results_default.csv`
- `results/summary_results_default.csv`
- `results/wilcoxon_tests_default.csv`
- `results/figures/{project}_f1_boxplot_default.png (5 files)`

Expected Results

Due to the use of `random_state=run` in the train/test splits and fixed seeds on the stochastic classifiers, results should be exactly reproducible. In the weighted experiment, the summary table should show Logistic Regression as the best-performing classifier on 4 of 5 projects by F1 score, with SVM winning on TensorFlow. SVM, Logistic Regression, and XGBoost should show statistically significant improvement over the Naive Bayes baseline ($p < 0.05$) on all projects.

Approximate expected F1 scores (weighted experiment):

Project	NB (Baseline)	SVM	Log. Regression	XGBoost
TensorFlow	0.036	0.707	0.685	0.672
PyTorch	0.023	0.508	0.537	0.439
Keras	0.092	0.642	0.650	0.608
MXNet	0.049	0.523	0.603	0.461
Caffe	0.016	0.273	0.360	0.339

Random Forest results are omitted from this table as it largely failed on this task (near-zero F1 on most projects). Full results including Random Forest are in `results/raw_results.csv` and `results/summary_results.csv`. The default experiment produces its own equivalent outputs with the `_default` suffix and will show substantially lower F1 scores on the improved classifiers — this is the intended contrast demonstrating the impact of class weighting.

Troubleshooting

- **NLTK download error:** Ensure internet access on first run. NLTK needs to download the stopwords corpus.
- **ModuleNotFoundError:** Ensure you activated the virtual environment and ran `pip install -r requirements.txt`.
- **FileNotFoundError on datasets:** Ensure you are running the script from inside the `lab1/` directory.